Chemistry - Graduate Level Assessment 7 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Calculate the heat transferred when 250 J of work is done on a system consisting of 1 mole of an ideal gas. At constant temperature, = 0 for the expansion of an ideal gas.
   (a) 1 J
   (b) 500 J
   (c) -250 J
   (d) -500 J
   (e) -100 J

2. Under which conditions will a real gas most closely behave as an ideal gas?
   (a) Low temperature and low pressure
   (b) Low temperature and high pressure
   (c) High volume and high pressure
   (d) High temperature and high pressure
   (e) High volume and low pressure

3. What is the value of log 0.0148?
   (a) 3.0000
   (b) -0.1703
   (c) -2.1703
   (d) -0.8297
   (e) -1.1703

4. Which of the following is classified as a conjugate acid-base pair?
   (a) HCl / NaOH
   (b) H+ / Cl-
   (c) H_3O+ / H_2O
   (d) O_2 / H_2O

5. How many ounces of silver alloy which is 28% silver must be mixed with 24 ounces of silver which is 8% silver to produce a new alloy which is 20% silver?
   (a) 48 ounces of silver
   (b) 42 ounces of silver
   (c) 54 ounces of silver
   (d) 72 ounces of silver
   (e) 60 ounces of silver

True / False (10 marks)

Answer True or False

6. True or False: The synthesis of n-butane is less efficient due to the competitive formation of ethane and hexane as side products.

7. True or False: Para-dichlorobenzene is commonly referred to as 1,2-dichlorobenzene, which is a structural isomer with differing properties.

8. True or False: The Boyle temperature of a van der Waals gas is expressed as T_B = [(2 a) / (Rb)].

9. True or False: The energy of a photon corresponding to the series limit of the Lyman series is 123.64 nm.

10. True or False: A silver dollar weighs approximately 26.73 grams, which is equivalent to 0.02673 kilograms and 26,730 milligrams.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the volume percentage of an E. coli culture at a limiting concentration of 10^9 cells per cm^3, given each cell's dimensions. Discuss the implications of this density on nutrient availability.

12. Discuss the net ionic equation that results from mixing solutions of NH 4 Br and AgNO 3. Analyze the ions involved and explain the significance of the reaction.

End of Paper